

CREDIT RISK ANALYSIS × CUSTOMER SEGMENTATION

An Explainable Machine-Learning Framework for Risk-Aware, Customer-Centric Banking Decisions

GERMAN CREDIT RISK + CUSTOMER PERSONALITY ANALYSIS | DATA • EDA • ML • SEGMENTATION • DECISION SUPPORT

MACHINE LEARNING

FINTech / BANKING

A3 • RESEARCH POSTER

01

ABSTRACT

Financial institutions need both reliable credit-risk assessment and a detailed understanding of customer behaviour. This project integrates supervised credit-risk classification with unsupervised customer segmentation. The credit branch classifies applicants as Good/Bad risk; the segmentation branch identifies natural behavioural groups. Both outputs are connected to explainable, actionable banking recommendations and an interactive Streamlit decision-support layer.

KEYWORDS

Credit Risk

Customer Segmentation

02

PROBLEM STATEMENT & OBJECTIVES

PROBLEM

- Manual decisions can be slow/inconsistent.
- Generic marketing ignores heterogeneous behaviour.
- High-risk applicants must be detected carefully.

OBJECTIVES

- Predict applicant risk.
- Discover customer groups.
- Predict future segment membership.
- Recommend targeted actions.

03

DATASETS & FEATURE GROUPS

GERMAN CREDIT RISK

Age • Sex • Job • Housing
Credit Amount • Duration
Savings Account • Checking Account
Purpose • other relevant credit attributes

TARGET: Good = Low Risk | Bad = High Risk

CUSTOMER PERSONALITY

Demographic: Age • Income • Education • Marital Status
Behaviour: Total Spending •
Web/Store/Catalog/Deals Purchases
Engagement: Website Visits • Campaign Responses • Recency
Financial: product spending + purchase frequency + value

04

PREPROCESSING & FEATURE ENGINEERING

CLEAN

Missing values • duplicates • outliers
ENCODE
Categorical → numerical
SPLIT

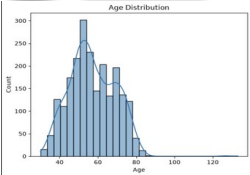
Training / testing for supervised learning

ENGINEER

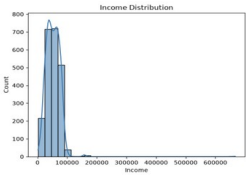
Age • Total Children • Total Spending
Customer Since • AcceptedAny • AgeGroup
TRANSFORM
Scaling / normalization / standardization

05

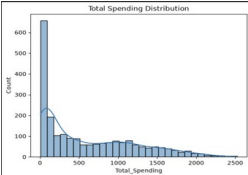
EDA: EVIDENCE FROM THE ANALYSIS



AGE



INCOME



SPENDING



SYSTEM ARCHITECTURE

RAW DATA → CLEANING → FEATURE ENGINEERING → MODELING → EXPLANATION → ACTION



CREDIT RISK

Decision Tree • Random Forest
Extra Trees • XGBoost

CUSTOMER SEGMENTATION

K-Means (primary)
Hierarchical • DBSCAN

EXPLAIN → PROFILE → RECOMMEND



END-TO-END METHODOLOGY



Risk evaluation prioritises high-risk recall; clustering evaluation focuses on compactness and separation.



MODEL CONFIGURATION & EVALUATION

CREDIT RISK

Decision Tree • Random Forest • Extra Trees •
XGBoost
Metrics: Accuracy • Precision • Recall • F1 • Confusion
Matrix • ROC-AUC

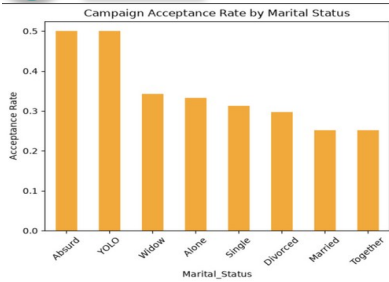
SEGMENTATION

K-Means • Hierarchical • DBSCAN
Validation: Elbow • Silhouette • Davies-Bouldin •
PCA

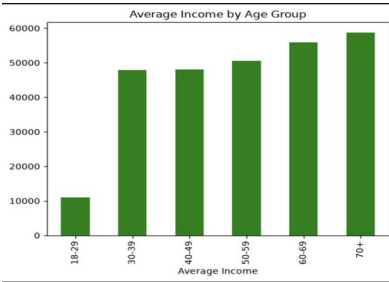
Select the best model using both statistical performance and business risk considerations.



SEGMENTATION VALIDATION & ACTUAL RESULTS



ELBOW METHOD



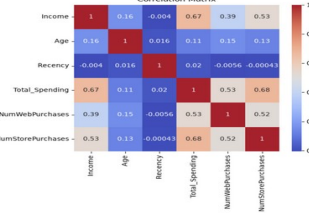
PCA — K=6

WCSS falls rapidly as K increases and then shows diminishing improvement around K=6-7. The implemented notebook uses K=6.

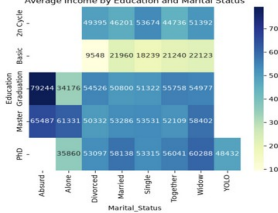
ID	N	INCOME	RECECY	SPEND	PROFILE
C0	298	44.7k	59	199	Older / low-spend
C1	309	76.5k	24	1171	Premium / high-value
C2	394	59.7k	46	894	Engaged / digital
C3	361	74.7k	71	1316	High-value / at-risk
C4	474	34.1k	23	120	Budget / low-spend
C5	380	32.0k	75	106	Low-engagement / at-risk



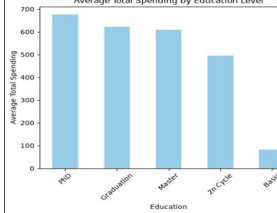
EDA — RELATIONSHIP STRUCTURE



CORRELATION



INCOME / AGE



CAMPAIGN RATE



WHAT THE EDA REVEALS

- Income ↔ Total Spending: $r \approx 0.67$
- Total Spending ↔ Store Purchases: $r \approx 0.68$
- Total Spending ↔ Web Purchases: $r \approx 0.53$
- Recency shows near-zero linear correlation with the selected value/purchase variables.
- Income and spending are strongly right-skewed; extreme observations need careful interpretation.
- Marital-status categories contain anomalies (e.g., Absurd / YOLO), highlighting the need for validation.



SEGMENT PROFILING → BANKING STRATEGY

PREMIUM	High income + high spending	Wealth management • premium banking • investment
GROWTH	Moderate income + increasing spending	Upselling • personalised offers
DIGITAL	High web purchases + online engagement	Digital products • mobile campaigns
PRICE-SENSITIVE	Low spending + frequent deals	Discounts • budget products
AT-RISK	Low engagement + high recency	Retention • re-engagement



CUSTOMER SEGMENT PREDICTION & STREAMLIT

NEW CUSTOMER
Profile + spending + purchases + engagement
↓
Random Forest / XGBoost / LightGBM
↓
Predicted Segment

INTERACTIVE APPLICATION
Input → preprocessing → inference
→ segment/risk → profile → recommendation
Real-time decision support



REAL-WORLD APPLICATIONS, LIMITATIONS & FUTURE WORK

APPLICATIONS
• Automated loan processing
• Risk-aware credit limits
• Digital lending / credit cards
• Targeted marketing & product recommendation
• Retention and financial inclusion

PRODUCTION REQUIREMENTS
• SHAP explainability
• Bias / fairness testing
• Privacy & security
• Regulatory compliance
• Monitoring + retraining
• Add trained credit-risk metrics